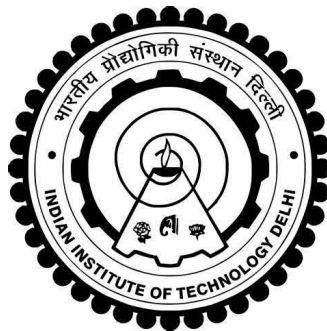


Indian Institute of Technology Delhi

Department of Mathematics



M.Tech. Project Final Report

Causal Learning for Strategic Policy Design (Based on NFHS-5)

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June 2026

Declaration

I certify that the work contained in this report has been carried out by me under the guidance of Prof. Rahul Singh and Prof. Biplab Paul, Department of Mathematics, Indian Institute of Technology Delhi. This work has not been submitted for the award of any other degree or diploma.

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Abstract

Population health policy in low- and middle-income countries requires moving beyond association to causation: identifying which modifiable indicators, when improved, produce the largest gains in health outcomes. We present a data-driven causal learning framework applied to the fifth round of India’s National Family Health Survey (NFHS-5), covering 131 standardised indicators across all 36 states and union territories.

The pipeline proceeds in four stages: (i) hierarchical clustering of 131 indicators into 32 thematically coherent groups using merge-height and within-cluster cohesion diagnostics; (ii) two-level PC-algorithm causal discovery (intra-cluster and inter-cluster meta-variable DAGs) with 100-resample bootstrap edge-confidence filtering; (iii) a three-criterion screening procedure (directional asymmetry $\Delta \geq 0.10$, undirected-edge fraction $\beta < 0.30$, partial-correlation residual $|\rho| \geq 0.35$) with empirically calibrated thresholds; and (iv) Average Treatment Effect (ATE) estimation along six validated pathways using four complementary estimators (IPW, DR Learner, T-Learner, X-Learner), with cross-estimator sign consistency as the primary evidence criterion.

All six pathways achieve pooled consistency: Family Planning (Any) reduces unmet spacing need (-3.22 pp), raises neonatal tetanus protection ($+5.55$ pp), and is linked to a C-section pathway via elevated blood pressure ($+9.71$ pp); first-trimester ANC raises facility-seeking for home births ($+1.70$ pp); women’s literacy increases mobile phone ownership ($+14.94$ pp); and men’s internet use is causally associated with men’s literacy ($+4.47$ pp). Stratified analysis reveals systematic effect heterogeneity: literacy effects on mobile ownership are nearly twice as large in rural settings and increase monotonically with income, indicating a demand-pull mechanism. These findings demonstrate that area- and income-stratified causal estimation should be the analytical default for NFHS-based policy design.

Keywords: Causal inference, NFHS-5, Public health policy, India, PC algorithm, Average Treatment Effect, Graph clustering, Stratified analysis, Causal structure learning, Machine learning, Doubly Robust estimation.

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1. Introduction

Population health in low- and middle-income countries is shaped by a dense web of socioeconomic, behavioural, and biological processes that interact in non-obvious ways. Policy interventions like investments in sanitation, family-planning programmes, nutrition campaigns are costly and politically constrained, so their expected effects must be estimated rigorously before deployment. Standard approaches in public health rely on observational associations drawn from large-scale surveys, yet association does not imply causation. A positive correlation between maternal literacy and child vaccination uptake may reflect a shared confounder (household wealth) rather than a direct effect of education; acting on spurious correlations wastes resources and may cause harm [11, 28].

Causal inference provides a principled framework for moving beyond association. Rather than asking *which indicators co-move*, it asks *what would happen if we intervened on a specific indicator*—a question that maps directly onto policy design. Prior work has applied causal methods to maternal and child health data: instrumental variable designs have estimated the effect of institutional delivery on neonatal mortality [20], propensity-score methods have examined skilled-birth attendance [4], and doubly-robust estimators have been deployed on Demographic and Health Survey data to study anaemia and stunting [18, 21]. However, most such analyses treat indicator pairs in isolation, ignoring the multi-dimensional dependency structure that governs how exposures and outcomes co-evolve across a national health system.

The empirical foundation of this study is the fifth round of India’s National Family Health Survey (NFHS-5), conducted during 2019–2021 [15], covering 131 standardised indicators (health, nutrition, family planning, women empowerment, education, etc.) across all 36 states and union territories. The resulting dataset is a high-dimensional observational system in which socioeconomic context, health behaviours, and biological outcomes are jointly recorded.

Our framework proceeds in four stages. *Structure discovery*: a correlation architecture is recovered, indicators are grouped into thematically coherent clusters via hierarchical clustering, and a two-level PC algorithm [34] with bootstrap validation identifies candidate directed pathways. *Causal screening*: three empirically calibrated criteria separate causal edges from confounding artefacts. *Causal effect estimation*: for each of six validated pathways, the ATE is estimated via four complementary estimators. *Heterogeneity analysis*: estimation is stratified by rural/urban area type and per-capita GSDP tertile.

1.1 Contributions

Our contributions are as follows:

1. We propose a scalable observational causal analysis pipeline combining graph-based indicator clustering with two-level constraint-based causal discovery, making PC-algorithm structure learning tractable on a 131-indicator national health survey.
2. We calibrate a principled three-criterion causal screening procedure (directional asymmetry, undirected-edge fraction, partial-correlation residual) using KDE trough, Jenks natural break, and F1-proxy analysis, providing data-justified thresholds rather than arbitrary cut-points.
3. We evaluate four complementary causal ML estimators (IPW, DR Learner, T-Learner, X-Learner) across six directed pathways inferred from NFHS-5, reporting 95% bootstrap confidence intervals and cross-estimator sign consistency.
4. We demonstrate that area- and income-stratified estimation substantially changes causal conclusions relative to pooled inference, and justify this choice via CATE heterogeneity analysis.
5. We provide a transparent five-stage pathway selection procedure from 88 bootstrap-screened edges to 6 final pathways, eliminating domain-implausible relationships.

1.2 Organisation of the Report

Chapter 2 reviews related work. Chapter 3 describes the dataset and preprocessing. Chapter 4 formalises the causal inference framework. Chapter 5 presents the graph clustering methodology. Chapter 6 describes the ATE estimation pipeline. Chapter 7 reports experimental results. Chapters 8 and 10 discuss findings and conclude. Appendices contain proofs, supplementary tables, and figures.

2. Related Work

2.1 Causal Inference in Population Health

Causal methods have been applied to health survey data with growing frequency. Lim et al. [20] used instrumental variables to estimate the effect of institutional delivery on neonatal mortality in South Asia. Bhutta et al. [4] applied propensity-score matching to skilled-birth attendance. Doubly-robust methods were used by Luque-Fernandez et al. [21] and Kennedy [18] on DHS data to study anaemia and stunting.

Constraint-based causal discovery algorithms, particularly the PC algorithm [34], have been applied to biomedical data to recover directed acyclic graphs from observational data under the faithfulness and Markov assumptions. Kalisch and Bühlmann [17] showed that PC scales to high-dimensional settings using ℓ_1 -penalised conditional independence tests. Bootstrap aggregation over PC outputs to quantify edge stability was formalised by Friedman and Koller [9].

Machine learning-augmented causal estimators have expanded estimation accuracy in small and moderate samples. The doubly robust DR Learner [7] achieves $\sqrt{n}$ -consistency if either the propensity or outcome model is correct. Meta-learner frameworks (T-Learner, X-Learner) were systematically compared by Künnel et al. [19], who showed that the X-Learner’s propensity-weighted imputation reduces variance under treatment group imbalance. Nie and Wager [26] showed that the R-Learner achieves minimax optimality on CATE estimation; the DR Learner is a special case. DoWhy [32], EconML [22], and CausalML [6] provide production-grade implementations.

CATE heterogeneity analysis provides direct empirical justification for stratified estimation [1, 42]. When CATE distributions differ between rural and urban strata, pooled ATE is a weighted average of signals that may even oppose in sign, a form of Simpson’s paradox [28].

2.2 NFHS and Indian Health Policy Analysis

The NFHS series is the principal source for population health statistics in India [15]. Previous NFHS-based studies have examined women’s empowerment [40] and contraceptive uptake [31], but typically treat indicator pairs in isolation without a formal causal framework. Economic development has been identified as a structural confounder in cross-state Indian health analyses [12, 36]. The mismatch between eco-

conomic ranking and health outcomes in Indian states [41] illustrates why causal rather than correlational analysis is essential.

2.3 Graph-Based Indicator Clustering

Community detection algorithms have been applied to correlation networks in economics and epidemiology. Louvain [5] and Leiden [39] maximise modularity on weighted graphs; spectral clustering via the graph Laplacian [25] provides a geometric alternative. To the best of our knowledge, no prior work has systematically swept and compared all four clustering paradigms on a health indicator network and selected the best partition for downstream causal PC search using a combined Silhouette–Modularity–NMI ranking.

3. Dataset and Preprocessing

3.1 NFHS-5 Survey

NFHS-5 was conducted during 2019–2021 [15] and covers 131 standardised indicators spanning fertility, nutrition, health behaviour, and social determinants across all 28 states and 8 union territories. The survey employed a stratified, two-stage sampling design with approximately 636,699 households interviewed, yielding data from 724,115 women aged 15–49 and 101,839 men aged 15–54. NFHS-5 reports each indicator separately for *Rural*, *Urban*, and *Total* strata, enabling area-stratified causal analysis without approximation.

3.2 Per-Capita GSDP Data

State-level per-capita Gross State Domestic Product (GSDP) for the NFHS-5 reference period was obtained from the Open Government Data Platform India [27]. States are assigned to income tertiles (*Low*, $\lesssim$ Rs. 90k; *Moderate*; *High*, $\gtrsim$ Rs. 180k per capita). Three union territories with missing GSDP are excluded, leaving $n = 33$ analysis units.

3.3 Preprocessing

1. **Name standardisation.** Inconsistent state spellings are harmonised.
2. **Missing-value handling.** Suppressed cells (“*”) are recoded to NaN; all 131 indicators are coerced to numeric; rows with $> 50\%$ missing values are dropped.
3. **Column selection.** Administrative identifiers are excluded; the 131 numeric indicator columns form the analytical feature set.
4. **Positivity check.** For each causal pathway, the complete-case subset must have $n \geq 8$ states.

Full details are in Appendix B.

4. Causal Structure Learning Framework

4.1 Structural Causal Models

A structural causal model (SCM) encodes the data-generating mechanism as

$$X_i := f_i(\text{Pa}_i, U_i), \quad i = 1, \dots, d,$$

where Pa_i denotes the parents of X_i in a directed acyclic graph (DAG) and U_i are mutually independent exogenous noise variables [28]. For a health outcome Y , exposure X , and covariates Z this specialises to $Y = f_Y(X, Z, U_Y)$, $X = f_X(Z, U_X)$.

4.2 Interventional Distributions and the Do-Operator

Observational summaries describe $E[Y \mid X = x]$, while causal reasoning concerns

$$E[Y \mid \text{do}(X = x)],$$

the expected outcome when X is externally set to x , removing all incoming edges to X in the DAG [28]. Under the backdoor criterion (Appendix A, Definition A.4):

$$P(Y \mid \text{do}(X = x)) = \sum_z P(Y \mid X = x, Z = z) P(Z = z).$$

4.3 Average Treatment Effect

Under consistency, conditional ignorability, and positivity (Appendix A), the ATE is identified as:

$$\text{ATE} = E[Y(1) - Y(0)] = E_Z[E[Y \mid X = 1, Z] - E[Y \mid X = 0, Z]]. \quad (4.1)$$

4.4 PC Algorithm for Causal Discovery

The PC algorithm [34] recovers the DAG skeleton by iteratively testing conditional independence $X_i \perp\!\!\!\perp X_j \mid S$, removing edges when independence holds, then orienting remaining edges through v-structure rules. Because complexity scales as $O(n^k)$ in the number of variables n and maximum degree k [17], applying PC directly to all 131

NFHS indicators is computationally intractable, motivating the two-level architecture described in Chapter 6.

5. Graph Clustering Methodology

5.1 Correlation Distance and Rank Structure

A correlation-distance matrix $D_{ij} = 1 - |\rho_{ij}|$ embeds indicators in a similarity space. Rank plots of pairwise correlations across overall, area-stratified, income-stratified, and partial-correlation stratifications (Appendix E) reveal: (i) right-skewed distributions with ~ 10 pairs exceeding $|r| = 0.70$; (ii) rural correlations are stronger than urban at every rank; (iii) GSDP suppresses rather than inflates associations.

Preliminary mediation analysis (Appendix F) confirmed that per-capita GSDP is a weak direct confounder (proportion mediated = -2.14%); women’s literacy is the dominant mediating channel and is adopted alongside GSDP as the joint confounder set $\mathbf{Z}$.

5.2 Hierarchical Clustering and Cluster-Count Selection

Average-linkage agglomerative clustering minimises inter-cluster distance,

$$d(A, B) = \frac{1}{|A||B|} \sum_{i \in A} \sum_{j \in B} D_{ij},$$

producing dendrograms aligned with recognisable development domains [23]. The number of clusters k is selected via two evidence-based diagnostics (Figure 5.1):

- **Merge-height elbow.** Large jumps between consecutive merge heights signal natural cut-points [38].
- **Within-cluster cohesion.** The elbow in $\bar{D}(k)$ marks diminishing returns [37].

Both diagnostics converge on $k \in [28, 36]$; $k = 32$ is chosen as the knee of the cohesion curve with mean intra-cluster distance $\bar{D} < 0.30$.

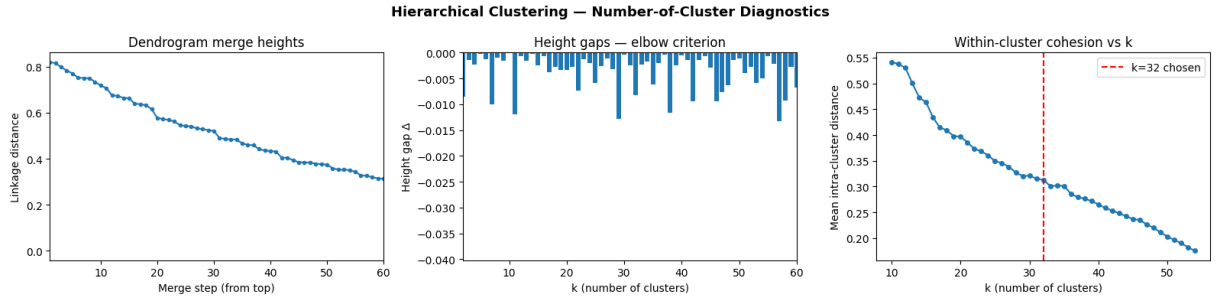


Figure 5.1: Hierarchical clustering diagnostics: (left) dendrogram merge heights, (centre) first-difference height gaps, (right) within-cluster cohesion curve. Red dashed line marks the chosen $k = 32$.

5.3 Graph-Based Clustering

Indicators form a weighted correlation network $G = (V, E, W)$ with $w_{ij} = |\rho_{ij}|$. Community detection maximises modularity [5]:

$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j). \quad (5.1)$$

Resolution sweeps over $\gamma \in \{0.5, \dots, 6.0\}$ select $\gamma = 3.0$ for both Louvain and Leiden (silhouette peak; aligned with hierarchical $k = 32$). Spectral clustering [25] uses $k = 28$ (local silhouette maximum = 0.2016).

5.4 Clustering Method Selection

Four candidate partitions are evaluated on Silhouette, Modularity Q , and mean pairwise NMI [35] (Table 5.1). Each method is ranked on each metric; the combined rank (lower = better) determines the method used for PC algorithm input.

Table 5.1: Clustering evaluation. Combined rank (sum of per-metric ranks; lower is better) determines the method used as input to the PC algorithm.

Method	Silhouette	Modularity	Mean NMI	Combined rank
Hierarchical	0.2541	0.3945	0.6968	4
Leiden	0.2096	0.3709	0.7273	5
Spectral	0.2016	0.2586	0.6850	9
Louvain	−0.1660	0.0753	0.6070	12

Hierarchical clustering achieves the highest silhouette and lowest combined rank and is selected for the PC algorithm.

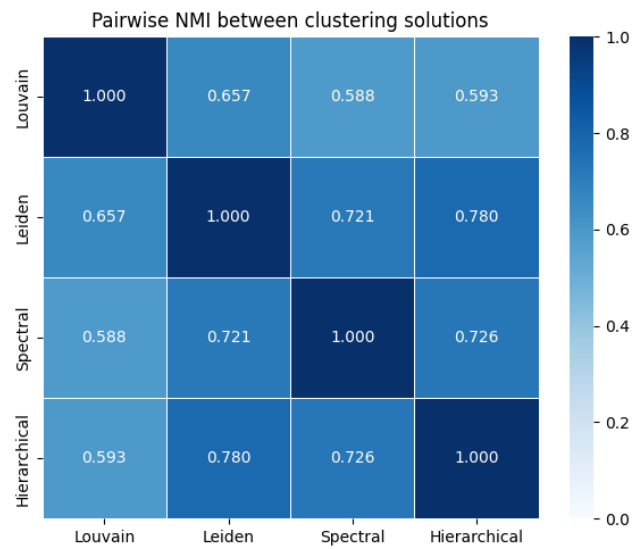


Figure 5.2: Pairwise NMI heatmap between four clustering solutions. High Leiden–Hierarchical (0.780) and Spectral–Hierarchical (0.726) NMI confirm consensus on the latent partition.

6. ATE Estimation Pipeline

6.1 Two-Level Causal Discovery Architecture

A two-level DAG architecture manages the $O(n^k)$ complexity of the PC algorithm while preserving cross-domain causal relationships. **Level 1 (intra-cluster):** the PC algorithm is applied within each of the 32 hierarchical clusters independently, recovering intra-domain directed edges. **Level 2 (inter-cluster):** each cluster C_k is summarised by a meta-variable $Z_k = |C_k|^{-1} \sum_{X_i \in C_k} X_i$; the PC algorithm is then applied on the 32 meta-variables to recover cross-domain directed pathways.

This decomposition is theoretically justified by the local Markov property of DAGs [8, 34]: if within-cluster indicators satisfy the cluster-level Markov condition, then the inter-cluster meta-variable DAG captures the essential cross-domain causal structure without running PC on all 131 indicators simultaneously. Clustering reduces each PC run to at most $|\text{cluster}| \leq 6$ variables ($\binom{6}{2} = 15$ pairs); the meta-variable run operates on only 32 nodes.

6.2 Bootstrap Edge Screening and Three-Criterion Causal Filtering

Bootstrap Aggregation

The PC algorithm is applied over $B = 100$ resamples (with replacement) of the n states; $\hat{P}(i \rightarrow j) = B^{-1} \sum_{b=1}^B \mathbf{1}\{i \rightarrow j \in G^{(b)}\}$ [9]. The 88 collected edge-direction pairs enter a three-stage filtering process.

Threshold Calibration

The three screening thresholds were calibrated empirically using distributional analysis (KDE and Jenks natural breaks [16]) and sensitivity-based analyses (Appendix C). The final values, $\Delta = 0.10$, $\beta < 0.30$, and $|\rho| \geq 0.35$, correspond to stable regions of the calibration curves and retain 43 of 88 candidate edge-direction pairs.

Three-Criterion Causal Screening

1. **Directional asymmetry ≥ 0.10 :** $\alpha_{AB} = \hat{P}(A \rightarrow B) - \hat{P}(B \rightarrow A) \geq 0.10$. Edges with $|\alpha_{AB}| < 0.10$ are orientation-ambiguous [34].

2. **Undirected-edge fraction** < 0.30 : Edges appearing undirected in $\geq 30\%$ of resamples lack sufficient orientation evidence.
3. **Partial-correlation residual** $|\rho_{AB \cdot Z}| \geq 0.35$: After regressing out GSDP and women’s literacy, the residual association must survive [28].

Of 88 bootstrap edges, 43 pass all three criteria; 18 are further removed by the domain-plausibility filter, leaving 25 final candidates.

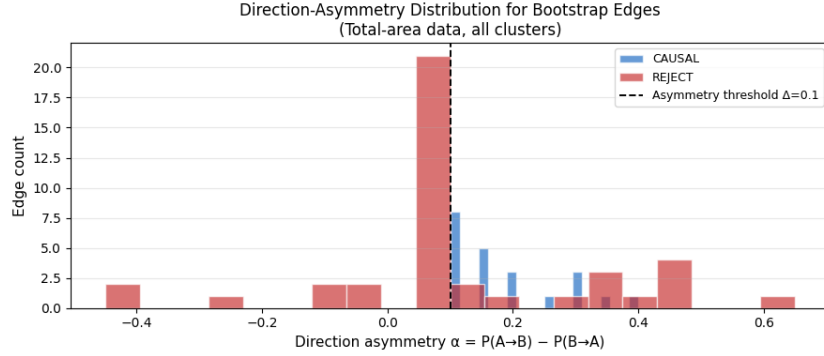


Figure 6.1: Direction-asymmetry distribution for all bootstrap edges. Blue: causal (passed all three criteria); red: rejected. Rejected edges concentrate near $\alpha \approx 0.1$, consistent with bidirectional confounding.

6.3 Pathway Selection: From 88 Edges to 6 Final Pathways

Six final pathways were obtained through five sequential stages:

Stage 1 – Bootstrap PC discovery. The PC algorithm is run over $B = 100$ resamples across all 32 clusters and the 32 meta-variables. A total of 88 bootstrap edge-direction pairs are collected.

Stage 2 – Three-criterion screening. Applying the empirically calibrated thresholds reduces the set from 88 to 43 edges.

Stage 3 – Domain-plausibility filtering. Eighteen edges are removed: (i) severity pairs, (ii) gender mirrors, (iii) temporally implausible directions, and (iv) demographic count variables acting as purported causes of clinical outcomes. This leaves 25 candidates.

Stage 4 – Candidate pool of 14. Fourteen pathways are retained as candidates after inspection over 4–5 independent pipeline runs. The 14 candidates are listed in Appendix D.

Stage 5 – Final 6 by estimator and stratification consistency. The four-estimator ATE pipeline is applied to all 14 candidates. The six pathways reported here

achieve: (a) consistent sign agreement ($\geq 3/4$ estimators) under pooled estimation; and (b) consistent sign agreement under at least two of the three area strata and two of the three income strata.

The six final pathways are:

1. Family Planning (Any) $\rightarrow$ Unmet Need for Spacing.
2. Family Planning (Any) $\rightarrow$ Neonatal Tetanus Protection.
3. ANC (First Trimester) $\rightarrow$ Home Births Taken to Facility.
4. Women Elevated Blood Pressure $\rightarrow$ C-Section (Public Facility).
5. Women Literacy $\rightarrow$ Mobile Phone Ownership.
6. Men Internet Use $\rightarrow$ Men Literacy.

Per-capita GSDP and women's literacy serve as confounders $\mathbf{Z}$ throughout. Treatment indicators $T \in \{0, 1\}$ are binarised at the state-level median.

6.4 Choice of Four Estimators

The four estimators span the space of modelling assumptions so that their agreement constitutes stronger evidence than any single estimator alone [7, 19?].

Inverse Probability Weighting (IPW)

Relies solely on the propensity model $\hat{e}(\mathbf{z}) = \hat{P}(T = 1 \mid \mathbf{Z} = \mathbf{z})$ [30]:

$$\widehat{\text{ATE}}_{\text{IPW}} = \frac{1}{n} \sum_{i=1}^n \left(\frac{T_i Y_i}{\hat{e}(\mathbf{z}_i)} - \frac{(1 - T_i) Y_i}{1 - \hat{e}(\mathbf{z}_i)} \right).$$

Implemented via DoWhy [32]; 200 bootstrap resamples.

Doubly Robust (DR) Learner

Augments IPW with an outcome model $\hat{\mu}_t(\mathbf{z})$, consistent if *either* model is correctly specified [7, 29]:

$$\widehat{\text{ATE}}_{\text{DR}} = \frac{1}{n} \sum_i \left[\hat{\mu}_1(\mathbf{z}_i) - \hat{\mu}_0(\mathbf{z}_i) + \frac{T_i(Y_i - \hat{\mu}_1(\mathbf{z}_i))}{\hat{e}(\mathbf{z}_i)} - \frac{(1 - T_i)(Y_i - \hat{\mu}_0(\mathbf{z}_i))}{1 - \hat{e}(\mathbf{z}_i)} \right].$$

Implemented via EconML LinearDRLearner [22] with $k = 3$ cross-fitting.

T-Learner

Fits separate outcome models for treated and control groups [19]: $\hat{\tau}(\mathbf{z}) = \hat{\mu}_1(\mathbf{z}) - \hat{\mu}_0(\mathbf{z})$. Implemented via CausalML [6].

X-Learner

Refines T-Learner via cross-group imputation [19]: $\hat{\tau}(\mathbf{z}) = \hat{e}(\mathbf{z}) \hat{\tau}_0(\mathbf{z}) + (1 - \hat{e}(\mathbf{z})) \hat{\tau}_1(\mathbf{z})$. Implemented via CausalML; 200 bootstrap resamples.

Consistency criterion. An ATE is classified as *Consistent* if at least 3 of 4 estimators agree on sign; otherwise *Partial*. Median ATE across estimators is the summary point estimate.

6.5 Stratified Estimation via CATE Heterogeneity

Pooled ATEs implicitly assume effect homogeneity across the rural–urban continuum and income levels, an assumption almost certainly violated in the Indian health context [11, 13]. State-level CATE distributions (Section 7.2, Figure 7.2) from the DR Learner and X-Learner reveal non-trivial variance across states for all pathways, providing direct empirical justification for stratified estimation [1, 42]. The full four-estimator pipeline is therefore re-run on (i) NFHS-5 Rural and Urban rows separately and (ii) the three per-capita GSDP tertile subsets.

7. Experimental Results

7.1 Pooled ATE Results

Full per-estimator ATEs with 95% confidence intervals are in Appendix I (Table I.1 and Table I.2). All six pathways achieve pooled cross-estimator sign consistency.

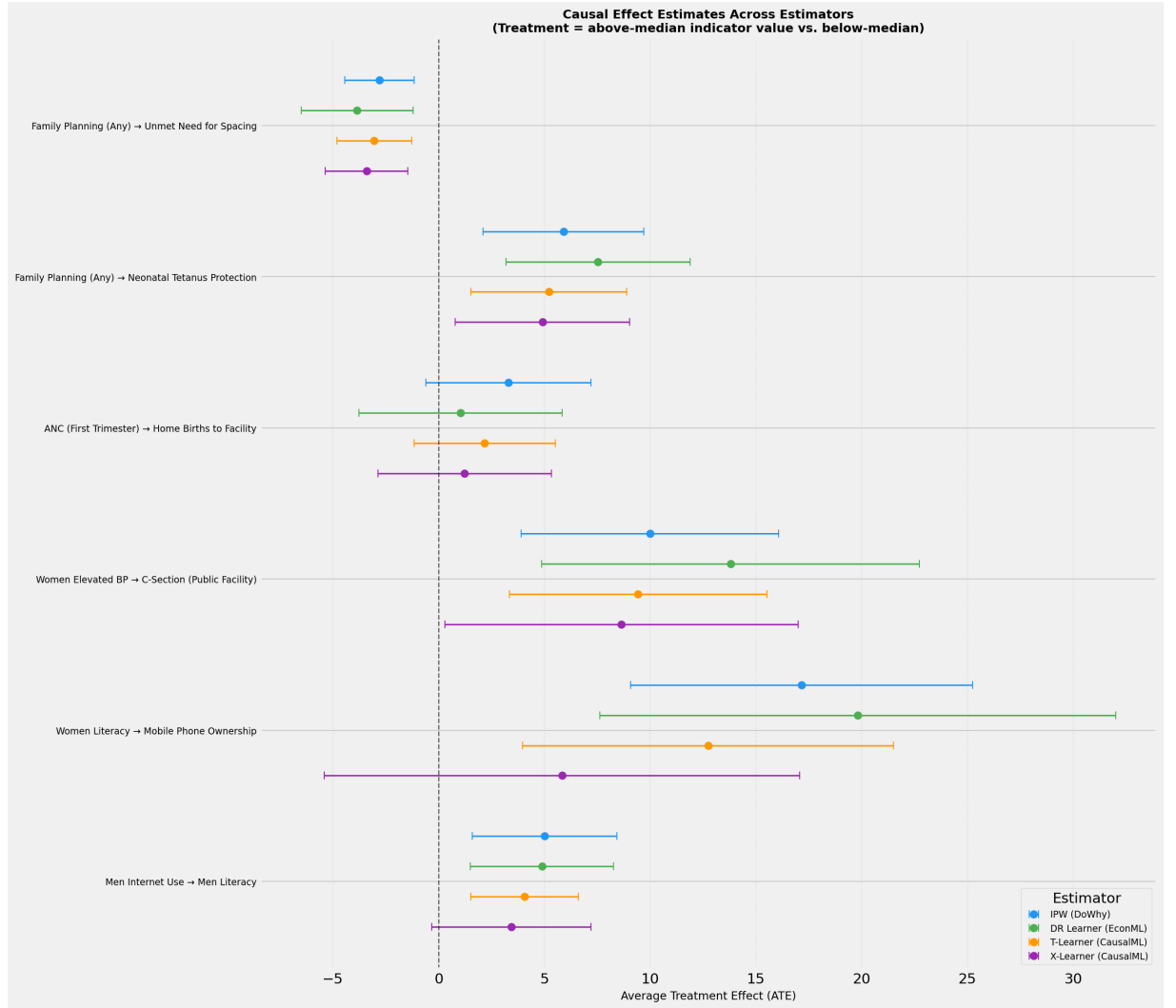


Figure 7.1: Forest plot of pooled ATE estimates with 95% CIs across all four estimators for each of the six pathways. Dashed line marks zero effect.

Family Planning (Any) → Unmet Need for Spacing (Median ATE = -3.22 pp, Consistent). The tightest cross-estimator agreement: IPW -2.81 , DR -3.85 , T -3.04 , X -3.41 ; all four achieve $p \leq 0.004$. Higher contraceptive prevalence mechanically reduces the fraction of women in the desired-but-unachieved spacing gap

[31].

Family Planning (Any) $\rightarrow$ Neonatal Tetanus Protection (Median ATE = +5.55 pp, Consistent). IPW +5.90 ($p < 0.001$), DR +7.53 ($p = 0.001$), T +5.21 ($p = 0.006$), X +4.90 ($p = 0.020$). FP system contact provides TT immunisation opportunities during reproductive-health visits [3].

ANC (First Trimester) $\rightarrow$ Home Births Taken to Facility (Median ATE = +1.70 pp, Consistent). IPW +3.30, DR +1.04, T +2.17, X +1.22; all positive. First-trimester ANC builds health-system trust, increasing postnatal facility care-seeking [15].

Women Elevated BP $\rightarrow$ C-Section (Public Facility) (Median ATE = +9.71 pp, Consistent). IPW +9.99 ($p < 0.001$), DR +13.79 ($p = 0.002$), T +9.42 ($p = 0.002$), X +8.64 ($p = 0.042$). Hypertensive disorders of pregnancy are a leading clinical indication for caesarean section [43].

Women Literacy $\rightarrow$ Mobile Phone Ownership (Median ATE = +14.94 pp, Consistent). IPW +17.15 ($p = 0.001$), DR +19.81 ($p = 0.001$), T +12.73 ($p = 0.004$); X +5.84 ($p = 0.309$) is the sole non-significant estimate, reflecting over-regularisation at $n = 28$. Literacy enables independent mobile phone access [10].

Men Internet Use $\rightarrow$ Men Literacy (Median ATE = +4.47 pp, Consistent). IPW +5.01, DR +4.88, T +4.06; X +3.44 is marginal. The PC algorithm consistently oriented this edge in the internet $\rightarrow$ literacy direction; we note the conventional prior is reversed and treat this pathway as exploratory [2].

7.2 CATE Heterogeneity

DR Learner and X-Learner state-level CATE distributions (Figure 7.2) reveal non-trivial variance across states for all six pathways. The FP-spacing CATE sits uniformly below zero, confirming a universal direction. The literacy-mobile CATE is widely dispersed ($\approx +5$ to $+30$ pp), motivating stratified analysis.



Figure 7.2: State-level CATE distributions (violin plots) from DR Learner and X-Learner. Red dashed line at zero. Non-trivial heterogeneity justifies stratified estimation.

7.3 Area-Stratified Results

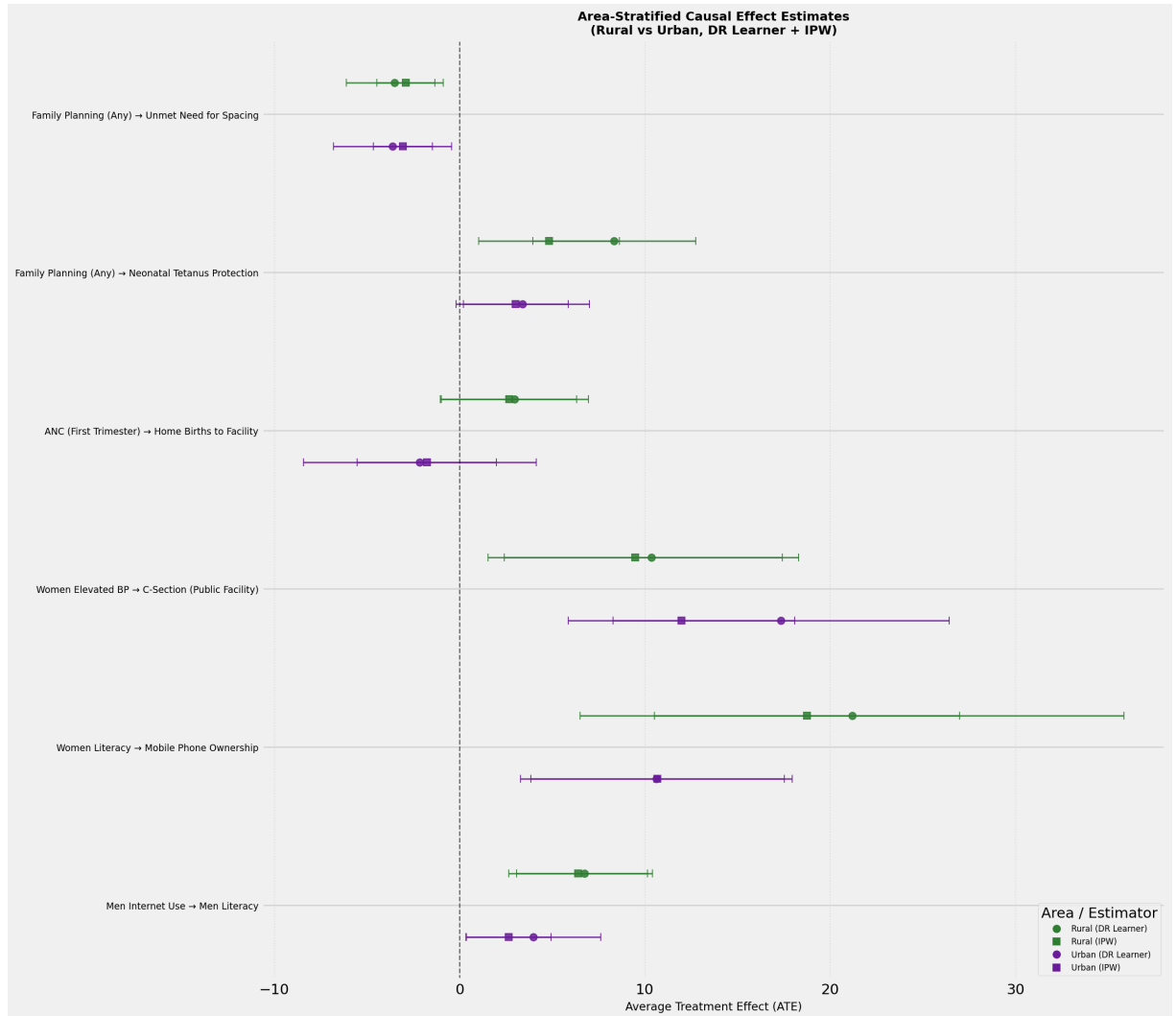


Figure 7.3: Area-stratified forest plot (DR Learner and IPW, 95% CIs). Green: Rural; purple: Urban.

Key findings (see Table I.3 in Appendix I):

- **FP → Spacing.** Rural -3.18 pp; Urban -3.06 pp (both Consistent). Cross-stratum agreement confirms a universal mechanism [31].
- **FP → Neonatal Tetanus.** Rural $+6.40$ pp; Urban $+4.03$ pp (both Consistent). The larger rural effect reflects greater relative PHC dependence.
- **ANC → Home Births.** Rural $+2.81$ pp (Consistent); Urban -1.65 pp (Partial). Urban facility-birth rates reduce the denominator, attenuating the effect.
- **Elevated BP → C-Section.** Rural $+7.78$ pp; Urban $+12.12$ pp (both Consistent). Larger urban effect reflects better-equipped urban facilities.

- **Literacy → Mobile Phone.** Rural +18.80 pp; Urban +10.34 pp (both Consistent). The rural effect is nearly twice the urban effect, consistent with literacy removing a binding constraint on mobile access [10].
- **Men Internet → Men Literacy.** Rural +5.74 pp; Urban +2.41 pp (both Consistent).

7.4 Income-Stratified Results

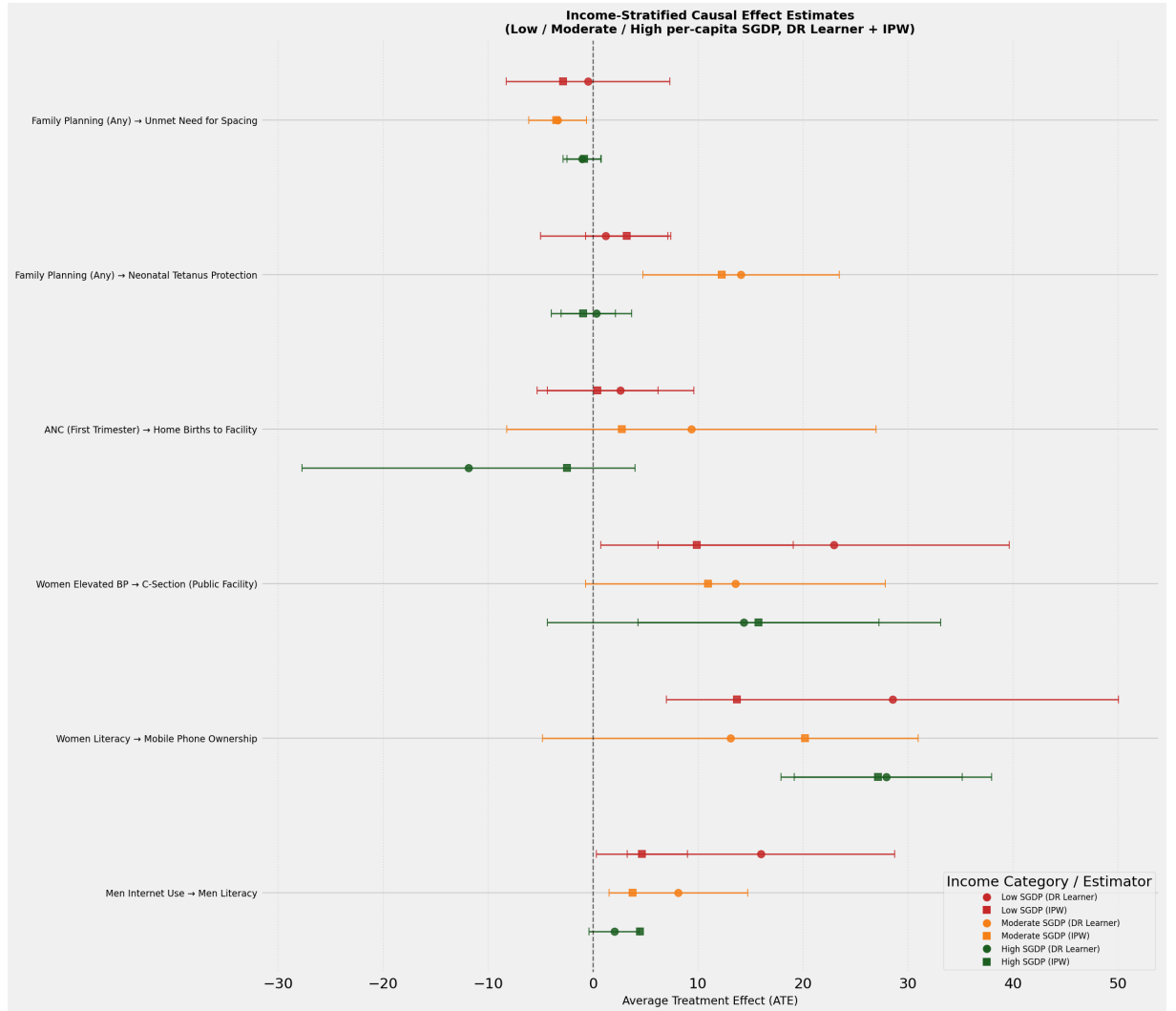


Figure 7.4: Income-stratified forest plot (DR Learner and IPW, 95% CIs). Red: Low GSDP; orange: Moderate; green: High.

Key findings (see Table I.4 in Appendix I):

- **FP → Spacing.** Monotone decline: Low -4.12 pp → Moderate -3.33 pp → High -0.79 pp (all Consistent). Diminishing marginal returns [14].

- **FP $\rightarrow$ Neonatal Tetanus.** Peaks in Moderate-GSDP (+12.46 pp); High-GSDP shows ceiling effect (Partial).
- **ANC $\rightarrow$ Home Births.** Peaks in Moderate (+5.11 pp); negative in High (-3.14 pp) due to near-full facility-birth coverage.
- **Elevated BP $\rightarrow$ C-Section.** Increases with income: Low +8.57 pp, Moderate +10.99 pp, High +12.42 pp (all Consistent), consistent with better surgical capacity.
- **Literacy $\rightarrow$ Mobile Phone.** Monotone increase: Low +13.81 pp, Moderate +16.26 pp, High +26.78 pp (all Consistent). High-GSDP mobile infrastructure amplifies the literacy demand effect.
- **Men Internet $\rightarrow$ Men Literacy.** Moderate-GSDP peak (+7.63 pp); Low and High approximately equal (+4.75 and +4.49 pp, all Consistent).

8. Discussion

This study demonstrates that combining data-driven causal discovery with stratified effect estimation yields policy-actionable insights that pooled observational analysis cannot provide.

8.1 Stratification as Default, Not Supplement

The systematic heterogeneity documented across all six pathways, confirmed by both CATE distributions and stratified ATEs, supports the conclusion that stratum-specific estimation should be the default for NFHS-based causal analyses. The literacy–mobile pathway effect is nearly twice as large in rural settings; the FP–tetanus pathway peaks in Moderate-GSDP states; the ANC effect reverses sign in high-GSDP states due to a ceiling effect. These are not noise: they are systematic, cross-estimator-consistent patterns with direct policy implications.

8.2 The Men Internet $\rightarrow$ Literacy Direction

The PC algorithm consistently oriented this edge as internet $\rightarrow$ literacy across multiple pipeline runs. A plausible mechanism is that states investing in digital literacy programmes (PMGDISHA, common service centres) drive both internet penetration and functional literacy simultaneously, with internet infrastructure as the leading indicator [2]. Nevertheless, reversing the edge (literacy $\rightarrow$ internet use) is the conventional prior; users of these results should treat this pathway as exploratory.

8.3 Cross-Estimator Consistency as Evidence Criterion

Because IPW relies solely on the propensity model, the DR Learner is doubly protected, and T/X-Learners use flexible outcome models, their convergence on a common direction provides evidence that findings are not artefacts of any single modelling assumption [7?].

8.4 Policy Implications

- (i) *FP programmes*: largest spacing effects in low-GSDP states; TT co-benefits require functional PHC system integration.

- (ii) *Women's literacy*: digital empowerment programmes targeting women should accompany literacy campaigns, particularly in rural and low-GSDP settings.
- (iii) *Hypertension screening*: the elevated BP $\rightarrow$ C-section pathway supports proactive antenatal BP screening.
- (iv) *Early ANC contact*: largest effect in Moderate-GSDP states where the health system is functional but home births remain common.

9. Limitations

1. **Small ecological sample.** The analysis operates at the state/UT level ($n \approx 28$ – 33 ; $n \approx 8$ – 10 within strata). Flexible ML nuisance models operate near capacity; CATE estimates are directionally informative rather than quantitatively precise. District-level microdata would allow larger, more reliable estimation.
2. **Conditional ignorability.** Unobserved state-level confounders (healthcare infrastructure density, governance quality) cannot be ruled out; all results should be interpreted as estimated treatment effects under standard identifiability assumptions.
3. **Ecological inference.** State-level associations cannot be directly attributed at the individual level.
4. **Binary treatment.** Binarising at the state median discards dose-response variation; continuous treatment methods would improve precision.
5. **PC stochasticity.** Results across 4–5 runs were used to stabilise pathway selection, but this cannot fully replace larger-sample confirmatory analysis.

10. Conclusion

We presented a data-driven causal learning framework combining evidence-based graph clustering, two-level bootstrap-PC causal discovery with empirically calibrated three-criterion screening, and multi-estimator ATE estimation on NFHS-5 data. Six pathways achieve consistent directional evidence across all four estimators: FP programmes reduce unmet spacing need (-3.22 pp) and raise neonatal tetanus protection ($+5.55$ pp); elevated BP among women is causally linked to C-section rates in public facilities ($+9.71$ pp); first-trimester ANC raises postnatal facility-seeking for home births ($+1.70$ pp); women’s literacy raises mobile phone ownership ($+14.94$ pp); and men’s internet use is associated with men’s literacy ($+4.47$ pp).

Stratified analysis revealed systematic heterogeneity: literacy effects on mobile ownership are twice as large in rural settings and increase monotonically with income; FP–tetanus effects peak in Moderate-GSDP states; BP–C-section effects scale with income. CATE distributions confirm non-trivial within-stratum heterogeneity, justifying stratified rather than pooled inference as the default for NFHS-based causal analyses.

Future work should extend the analysis to district-level NFHS microdata with continuous treatments, apply LiNGAM [33] for full DAG orientation, and exploit successive NFHS rounds for panel-based causal identification.

Code and Data Availability

The code for data preprocessing, causal structure learning, ATE estimation, and figure generation is publicly available at:

<https://github.com/abdheshdash8/Causal-Inference-for-health-and-nutrition-studies->

NFHS-5 data are publicly available at <https://dhsprogram.com/data>. Per-capita GSDP data are available from the Open Government Data Platform India.

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A. Causal Assumptions and Definitions

Definition A.1 (Consistency). *For any treatment level x : $X = x \Rightarrow Y = Y(x)$. This assumes a well-defined intervention and no interference between units [24].*

Definition A.2 (Conditional Ignorability). $(Y(1), Y(0)) \perp\!\!\!\perp X \mid Z$.

Definition A.3 (Positivity). *For all z with $P(Z = z) > 0$: $0 < P(X = 1 \mid Z = z) < 1$.*

Definition A.4 (Backdoor Criterion). *Z satisfies the backdoor criterion relative to (X, Y) if (i) no element of Z is a descendant of X and (ii) Z blocks all backdoor paths from X to Y . Under this condition: $P(Y \mid \text{do}(X = x)) = \sum_z P(Y \mid X = x, Z = z) P(Z = z)$.*

Proof: Backdoor Adjustment

Proof. Step 1: $P(Y \mid \text{do}(X = x)) = \sum_z P(Y \mid \text{do}(X = x), Z = z) P(Z = z \mid \text{do}(X = x))$. *Step 2:* Since Z has no descendant of X , $P(Z = z \mid \text{do}(X = x)) = P(Z = z)$. *Step 3:* By do-calculus Rule 2, $P(Y \mid \text{do}(X = x), Z = z) = P(Y \mid X = x, Z = z)$. *Step 4:* Substituting yields the backdoor formula. □ □

B. Dataset Construction and Scope

The raw NFHS-5 key-indicator file is merged with per-capita GSDP for 2020–21 [27]. Three union territories (Dadra & Nagar Haveli–Daman & Diu, Lakshadweep, Ladakh) are excluded due to missing GSDP, leaving $n = 33$ states/UTs for pooled analysis and $n \approx 28$ after pathway-specific complete-case filtering. Three design choices mitigate the small- n problem: (i) doubly-robust and X-Learner estimators are consistent under model misspecification; (ii) bootstrap CIs (200 resamples) propagate uncertainty non-parametrically; (iii) cross-estimator sign agreement filters inconclusive findings.

C. Threshold Calibration

The three screening criteria were calibrated empirically using the full set of 88 bootstrap edge-direction pairs. Figure C.1 summarises the calibration procedure.

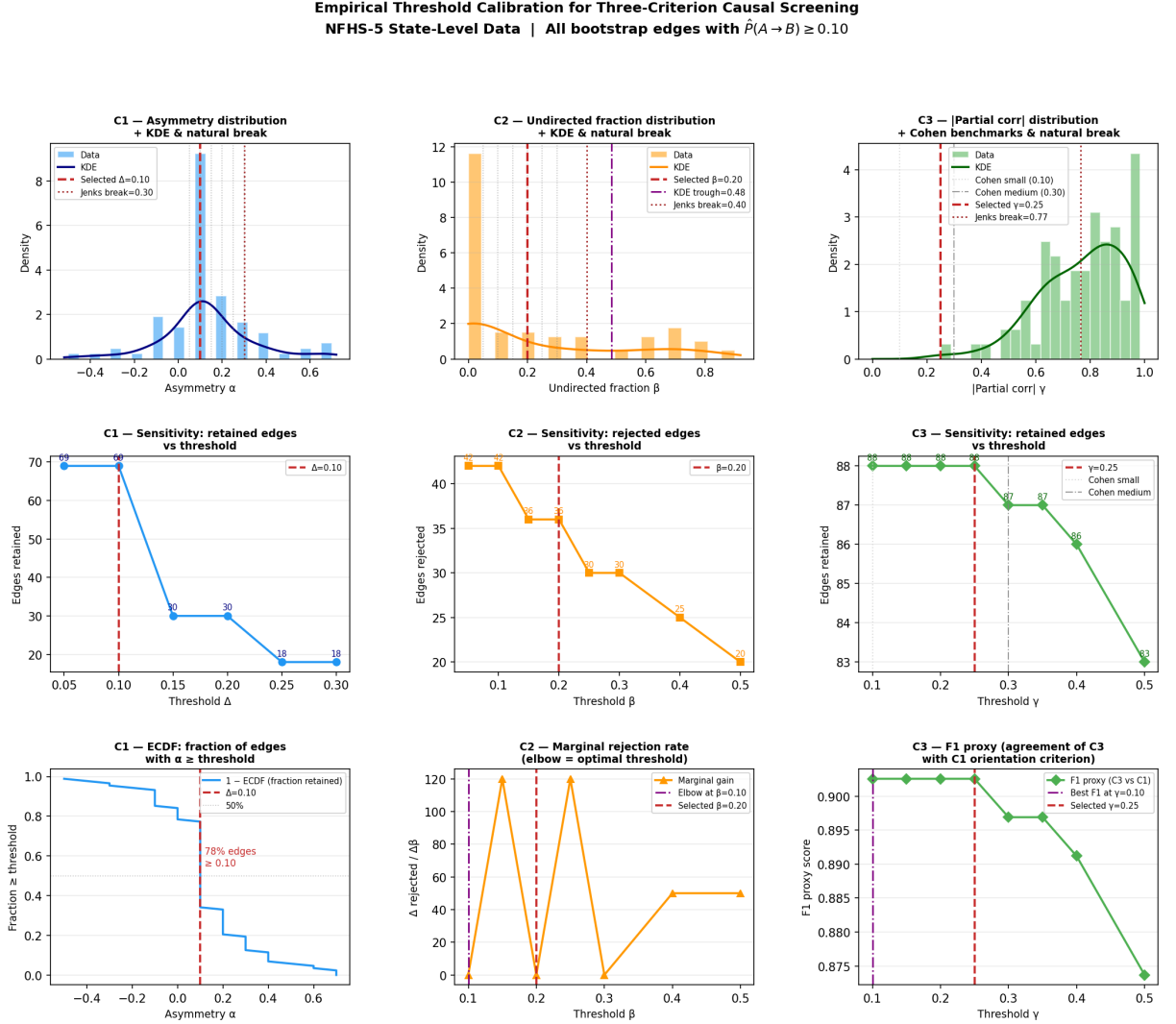


Figure C.1: Empirical calibration of the three screening criteria: directional asymmetry (Δ), undirected-edge fraction (β), and partial-correlation residual magnitude ($|\rho|$). Vertical dashed lines indicate the selected thresholds.

Criterion 1: Directional asymmetry (Δ). The asymmetry distribution is strongly concentrated near zero. The selected threshold $\Delta = 0.10$ coincides with the bootstrap confidence floor and retains approximately 80% of candidate edges, whereas larger thresholds lead to abrupt reductions without corresponding increase in directional certainty.

Criterion 2: Undirected-edge fraction (β). The distribution of β exhibits a clear separation between low-ambiguity and highly ambiguous edges. The KDE trough ($\beta \approx 0.47$) and Jenks break ($\beta \approx 0.40$) indicate a transition region beyond which orientations become unstable. Selected: $\beta < 0.30$.

Criterion 3: Partial-correlation residual ($|\rho|$). The F1-proxy remains nearly constant for thresholds between 0.10 and 0.40, while edge retention begins to decline more rapidly beyond $\gamma = 0.35$. Selected: $|\rho| \geq 0.35$.

D. Candidate Pathway Pool (14 Pathways)

Table D.1: 14 candidate pathways and selection outcome. “Pooled” = consistent across $\geq 3/4$ estimators in pooled analysis. “Selected” = also achieves stratification consistency.

#	Pathway	Pooled	Selected
1	Solid Food & BF $\rightarrow$ Diarrhoea (ORS)	Partial	No
2	FP(Any) $\rightarrow$ BCG Vaccination	✓	No
3	Pregnant Anaemia $\rightarrow$ Excl. Breastfeeding	✓	No
4	FP(Any) $\rightarrow$ Unmet Need for Spacing	✓	Yes
5	FP(Any) $\rightarrow$ Neonatal Tetanus Protection	✓	Yes
6	Physical Violence $\rightarrow$ HH Decision Participation	✓	No
7	ANC (First Trimester) $\rightarrow$ Home Births (Facility)	✓	Yes
8	Postnatal Care $\rightarrow$ Vitamin A Dose	Partial	No
9	Women Elevated BP $\rightarrow$ C-Section (Public)	✓	Yes
10	Women Tobacco $\rightarrow$ Children Overweight	Partial	No
11	Women Literacy $\rightarrow$ Mobile Phone Ownership	✓	Yes
12	FP(Any) $\rightarrow$ Diarrhoea (Taken to Facility)	Partial	No
13	Men Internet Use $\rightarrow$ Men Literacy	✓	Yes
14	Pregnant Anaemia $\rightarrow$ Child Anaemia (6–59m)	Partial	No

E. Exploratory Correlation Analysis

Indicator Distributions

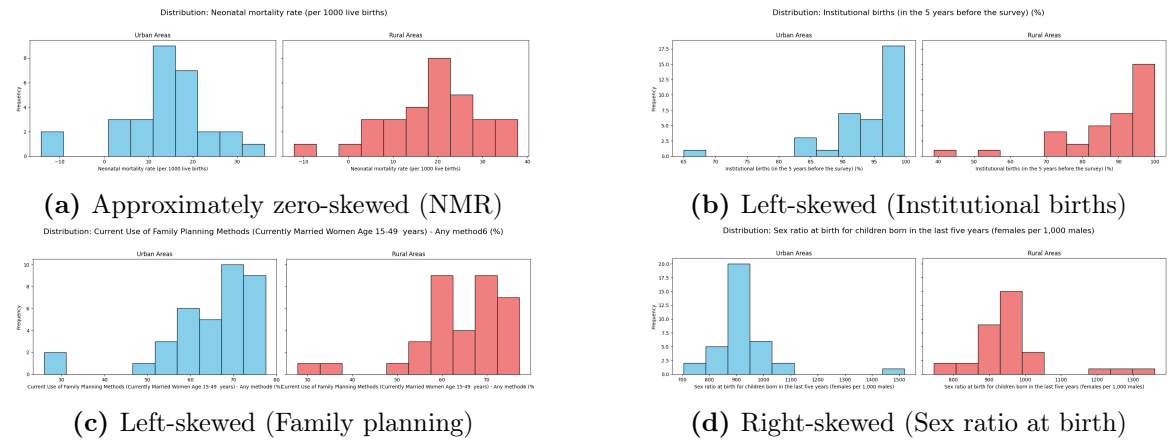


Figure E.1: Indicator-level distribution diagnostics.

Correlation Rank Plots

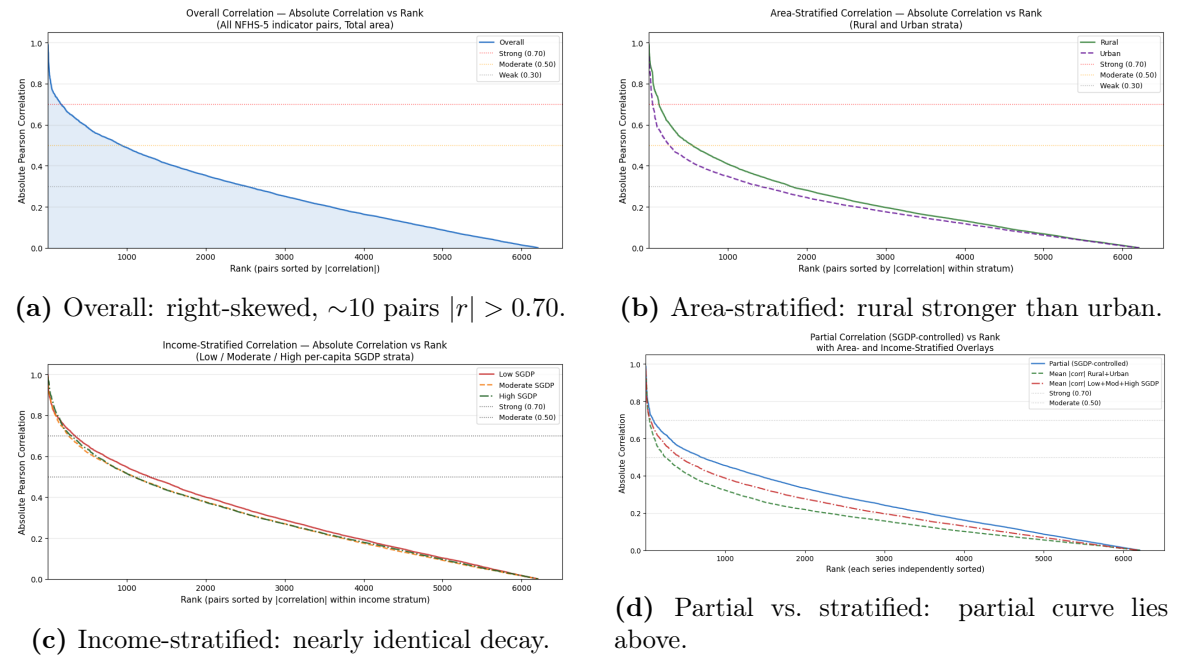


Figure E.2: Correlation rank plots across four stratifications.

F. Mediation Analysis

OLS models and mediation structural equations confirm that per-capita GSDP is a weak direct confounder (proportion mediated = -2.14%), while mediation through women’s literacy is significant. This motivated adopting women’s literacy alongside GSDP as the joint confounder set **Z**.

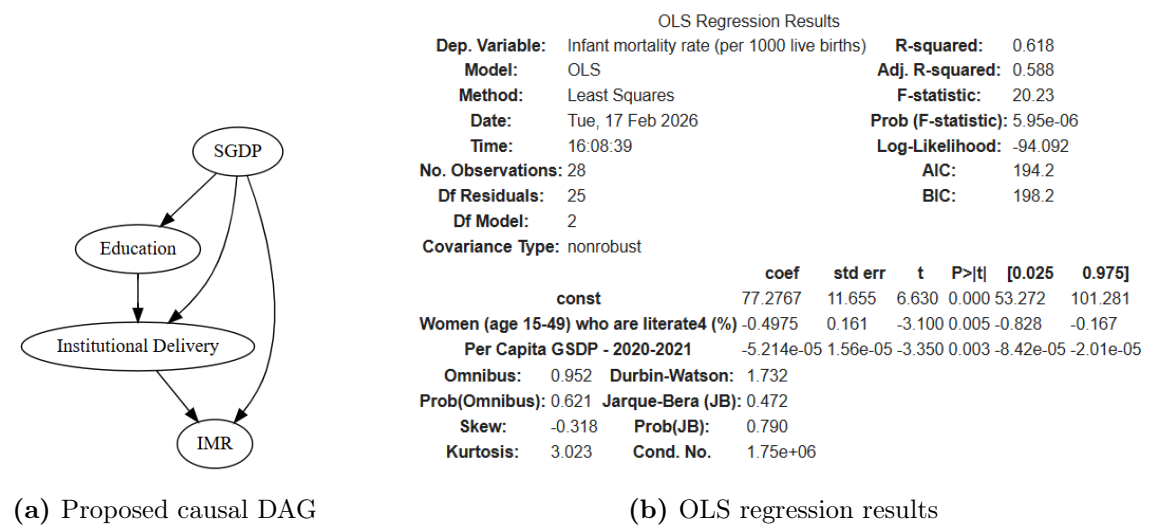


Figure F.1: Proposed DAG and OLS regression summary for mediation analysis.

G. Clustering Figures

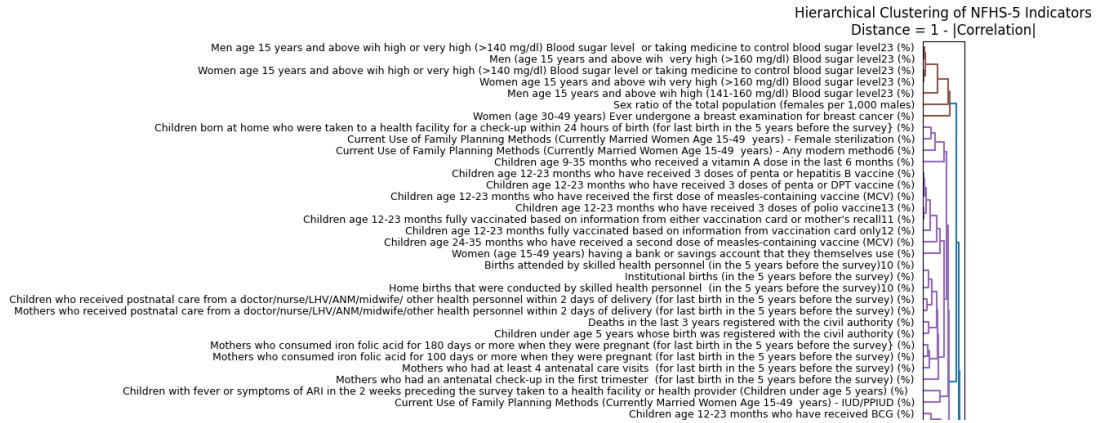


Figure G.1: Hierarchical clustering dendrogram ($k = 32$ cut shown).

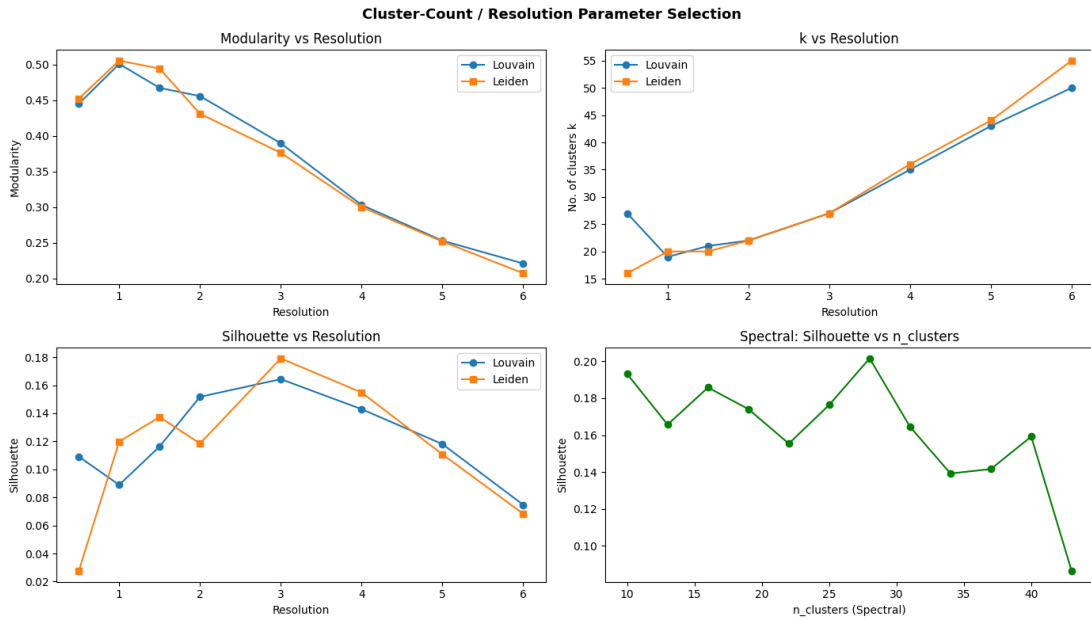


Figure G.2: Parameter sweep: Louvain/Leiden modularity and k vs. resolution (top); silhouette vs. resolution (bottom-left); Spectral silhouette vs. n clusters (bottom-right).

- Louvain Module 0
 - Women (age 15-49) with 10 or more years of schooling (%)
 - Institutional births in public facility (in the 5 years before the survey) (%)
 - Children age 12-23 months who received most of their vaccinations in a public health facility (%)
 - Children age 12-23 months who received most of their vaccinations in a private health facility (%)
- Louvain Module 1
 - Population living in households that use an improved sanitation facility2 (%)
 - Children under age 3 years breastfed within one hour of birth15 (%)
 - Women (age 15-49) who are literate4 (%)
 - Female population age 6 years and above who ever attended school (%)
 - Ever-married women age 18-49 years who have experienced physical violence during any pregnancy (%)
 - Ever-married women age 18-49 years who have ever experienced spousal violence27 (%)
 - Currently married women (age 15-49 years) who usually participate in three household decisions25 (%)
- Louvain Module 2
 - Children Prevalence of symptoms of acute respiratory infection (ARI) in the 2 weeks preceding the survey (Children under age 5 years) (%)
 - Prevalence of diarrhoea in the 2 weeks preceding the survey (Children under age 5 years) (%)
 - Population living in households with electricity (%)
 - Population below age 15 years (%)
 - Total Fertility Rate (number of children per woman)
 - Children under 5 years who are stunted (height-for-age)18 (%)

(a) Louvain ($\gamma = 3.0$)

- Leiden Module 0
 - Children under age 5 years whose birth was registered with the civil authority (%)
 - Deaths in the last 3 years registered with the civil authority (%)
 - Home births that were conducted by skilled health personnel (in the 5 years before the survey)10 (%)
 - Institutional births (in the 5 years before the survey) (%)
 - Mothers who received postnatal care from a doctor/nurse/LHW/ANM/midwife/other health personnel within 2 days of delivery (for last birth in the 5 years before the survey) (%)
 - Births attended by skilled health personnel (in the 5 years before the survey)10 (%)
 - Mothers who had an antenatal check-up in the first trimester (for last birth in the 5 years before the survey) (%)
 - Current Use of Family Planning Methods (Currently Married Women Age 15-49 years) - IUD/PPIUD (%)
 - Children with fever or symptoms of ARI in the 2 weeks preceding the survey taken to a health facility or health provider (Children under age 5 years) (%)
 - Children age 9-35 months who received a vitamin A dose in the last 6 months (%)
- Leiden Module 1
 - Mothers who consumed iron folic acid for 180 days or more when they were pregnant (for last birth in the 5 years before the survey) (%)
 - Mothers who consumed iron folic acid for 100 days or more when they were pregnant (for last birth in the 5 years before the survey) (%)
 - Births in a private health facility that were delivered by caesarean section (in the 5 years before the survey) (%)
 - Men age 15 years and above with high (141-160 mg/dl) Blood sugar level23 (%)
 - Births delivered by caesarean section (in the 5 years before the survey) (%)
 - Births in a public health facility that were delivered by caesarean section (in the 5 years before the survey) (%)
 - Men age 15 years and above with high or very high (>140 mg/dl) Blood sugar level or taking medicine to control blood sugar level23 (%)
 - Women age 15 years and above with high or very high (>140 mg/dl) Blood sugar level or taking medicine to control blood sugar level23 (%)
 - Women age 15 years and above with very high (>160 mg/dl) Blood sugar level23 (%)
 - Men (age 15 years and above with very high (>160 mg/dl) Blood sugar level23 (%)
- Leiden Module 2
 - Children age 12-23 months fully vaccinated based on information from vaccination card only12 (%)
 - Women (age 15-49 years) having a bank or savings account that they themselves use (%)
 - Children age 12-23 months who have received the first dose of measles-containing vaccine (MCV) (%)
 - Children age 24-35 months who have received a second dose of measles-containing vaccine (MCV) (%)
 - Children age 12-23 months who have received 3 doses of penta or hepatitis B vaccine (%)
 - Children age 12-23 months fully vaccinated based on information from either vaccination card or mother's recall11 (%)
 - Children age 12-23 months who have received 3 doses of penta or DPT vaccine (%)
 - Children age 12-23 months who have received 3 doses of polio vaccine13 (%)

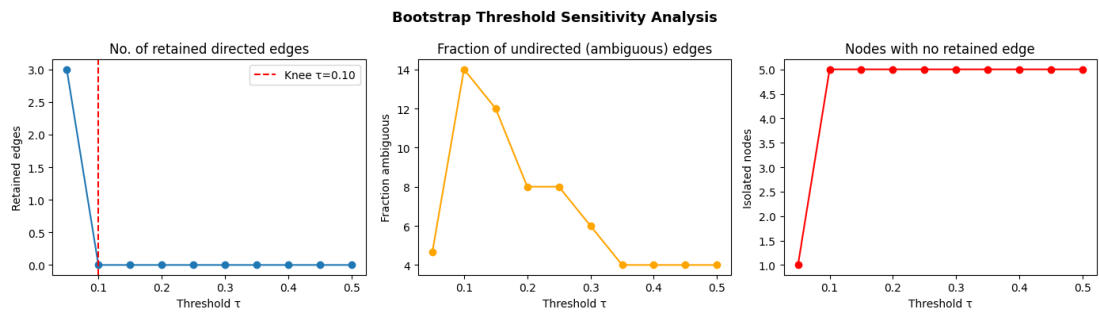
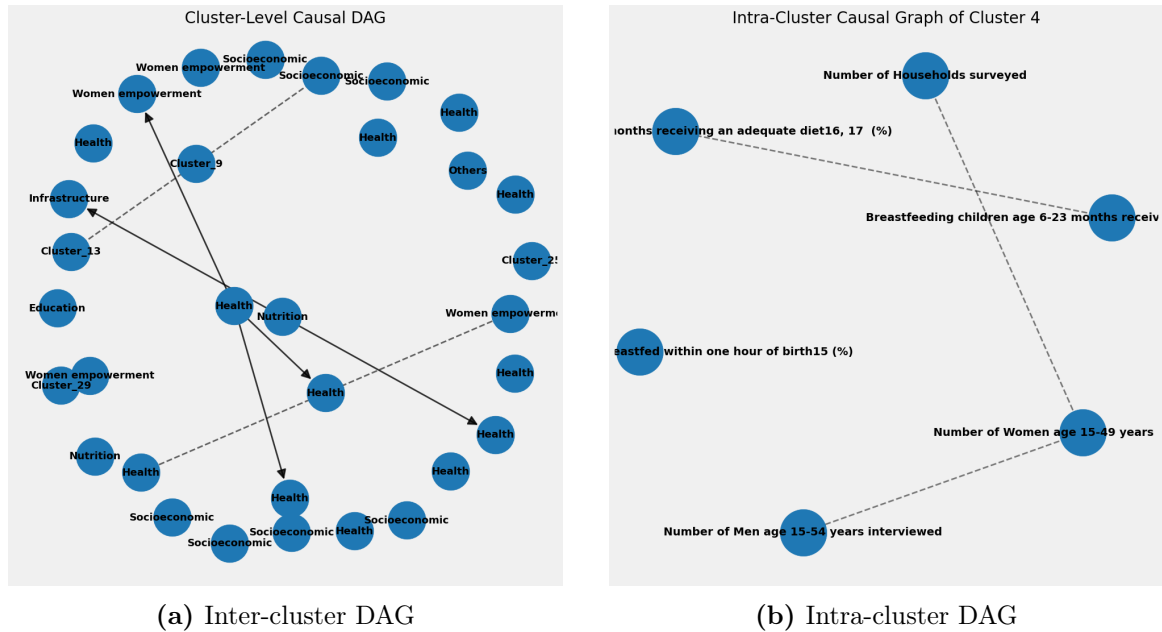
(b) Leiden ($\gamma = 3.0$)

- Spectral Cluster 0
 - Health worker ever talked to female non-users about family planning (%)
 - Current users ever told about side effects of current method of family planning8 (%)
- Spectral Cluster 1
 - Households using clean fuel for cooking3 (%)
 - Population living in households with an improved drinking-water source1 (%)
 - Men (age 15-49) who have ever used the internet (%)
 - Children under 5 years who are stunted (height-for-age)18 (%)
 - Women (age 15-49 years) who are overweight or obese (BMI ≥ 25.0 kg/m²)21 (%)
 - Men (age 15-49 years) who are overweight or obese (BMI ≥ 25.0 kg/m²) (%)
 - Women age 15-24 years who use hygienic methods of protection during their menstrual period26 (%)
 - Men (age 15-49 years) who know that consistent condom use can reduce the chance of getting HIV/AIDS (%)
 - Women (age 15-49 years) who know that consistent condom use can reduce the chance of getting HIV/AIDS (%)
 - Women (age 30-49 years) Ever undergone a screening test for cervical cancer (%)
- Spectral Cluster 2
 - Ever-married women age 18-49 years who have experienced physical violence during any pregnancy (%)
 - Population living in households that use an improved sanitation facility2 (%)
 - Ever-married women age 18-49 years who have ever experienced spousal violence27 (%)
- Spectral Cluster 3
 - Women (age 15-49) with 10 or more years of schooling (%)
 - Women age 15-19 years who were already mothers or pregnant at the time of the survey (%)
 - Adolescent fertility rate for women age 15-19 years5
 - Men (age 15-49) with 10 or more years of schooling (%)
 - Sex ratio at birth for children born in the last five years (females per 1,000 males)
 - Mothers who consumed iron folic acid for 180 days or more when they were pregnant (for last birth in the 5 years before the survey) (%)
 - Current Use of Family Planning Methods (Currently Married Women Age 15-49 years) - Pill (%)
 - Women age 15 years and above who use any kind of tobacco (%)
 - Women age 20-24 years married before age 18 years (%)
 - Men age 15 years and above who use any kind of tobacco (%)

(c) Spectral ($k = 28$)

Figure G.3: Graph-based clustering solutions.

H. Causal Discovery Figures



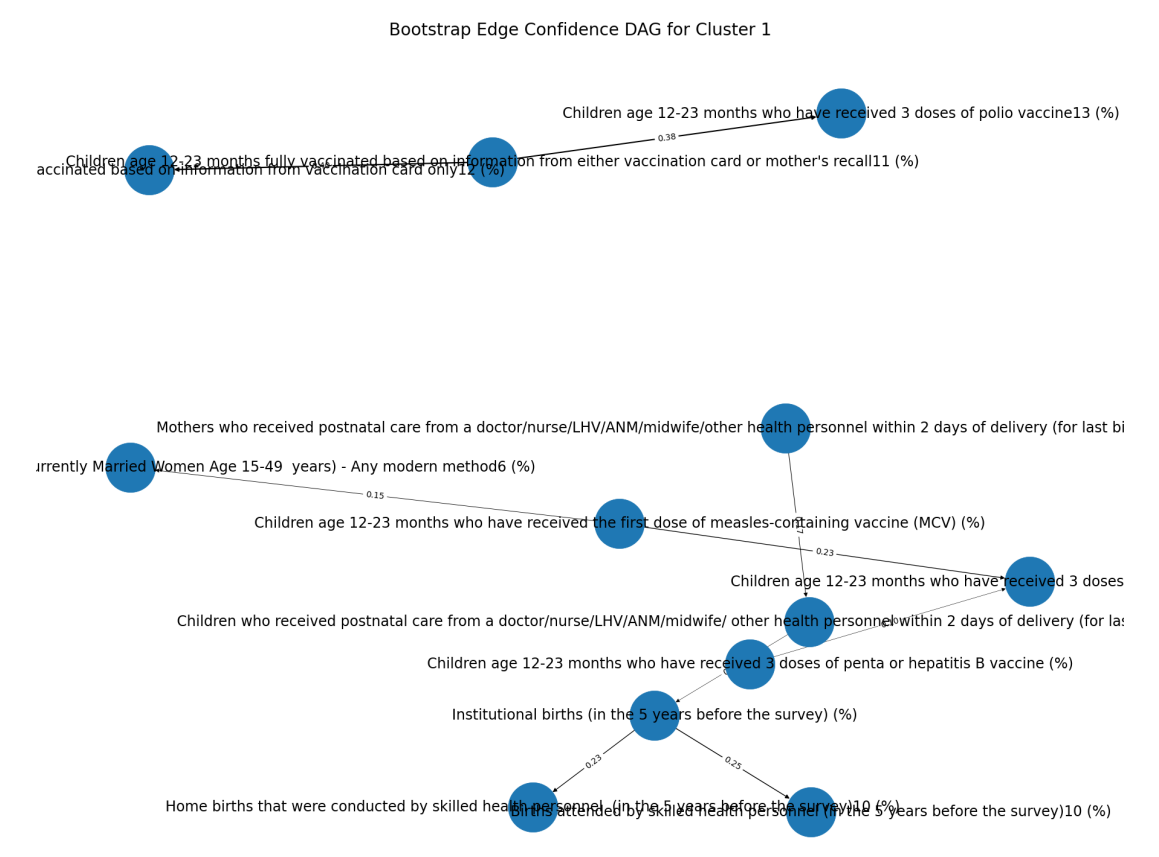


Figure H.3: Intra-cluster DAG for Cluster 1 with bootstrap edge confidence scores.

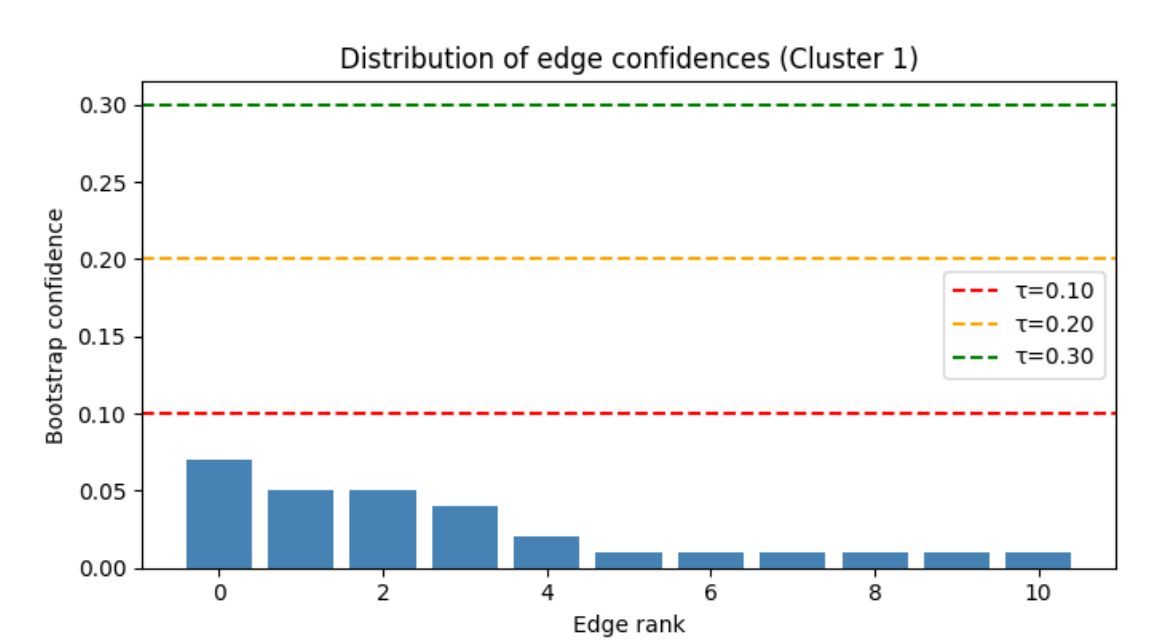


Figure H.4: Distribution of bootstrap confidence values for Cluster 1 directed edges.

I. Full ATE Tables and Heatmaps

Table I.1: Pooled ATE estimates (percentage points) for six causal pathways across four estimators ($n = 24$ – 28 states). A checkmark denotes $\geq 3/4$ estimators agree on sign.

Pathway	IPW	DR	T	X	Median	Cons.
FP $\rightarrow$ Unmet Spacing	−2.81	−3.85	−3.04	−3.41	−3.22	✓
FP $\rightarrow$ Neonatal Tetanus	+5.90	+7.53	+5.21	+4.90	+5.55	✓
ANC $\rightarrow$ Home Births	+3.30	+1.04	+2.17	+1.22	+1.70	✓
Elevated BP $\rightarrow$ C-Section	+9.99	+13.79	+9.42	+8.64	+9.71	✓
Literacy $\rightarrow$ Mobile Phone	+17.15	+19.81	+12.73	+5.84	+14.94	✓
Men Internet $\rightarrow$ Men Literacy	+5.01	+4.88	+4.06	+3.44	+4.47	✓

Table I.2: 95% confidence intervals for pooled ATE estimates.

Pathway	IPW	DR Learner	T-Learner	X-Learner
FP $\rightarrow$ Spacing	[−4.45, −1.17]	[−6.50, −1.20]	[−4.80, −1.29]	[−5.36, −1.45]
FP $\rightarrow$ Neonatal Tetanus	[3.26, 8.54]	[3.18, 11.89]	[1.52, 8.89]	[0.79, 9.02]
ANC $\rightarrow$ Home Births	[0.11, 6.49]	[−3.76, 5.84]	[−1.16, 5.50]	[−2.88, 5.33]
Elevated BP $\rightarrow$ C-Section	[5.41, 14.57]	[4.88, 22.71]	[3.34, 15.51]	[0.30, 16.98]
Literacy $\rightarrow$ Mobile Phone	[8.19, 26.12]	[7.61, 32.01]	[3.96, 21.51]	[−5.40, 17.07]
Men Internet $\rightarrow$ Literacy	[2.37, 7.64]	[1.50, 8.26]	[1.53, 6.59]	[−0.32, 7.20]

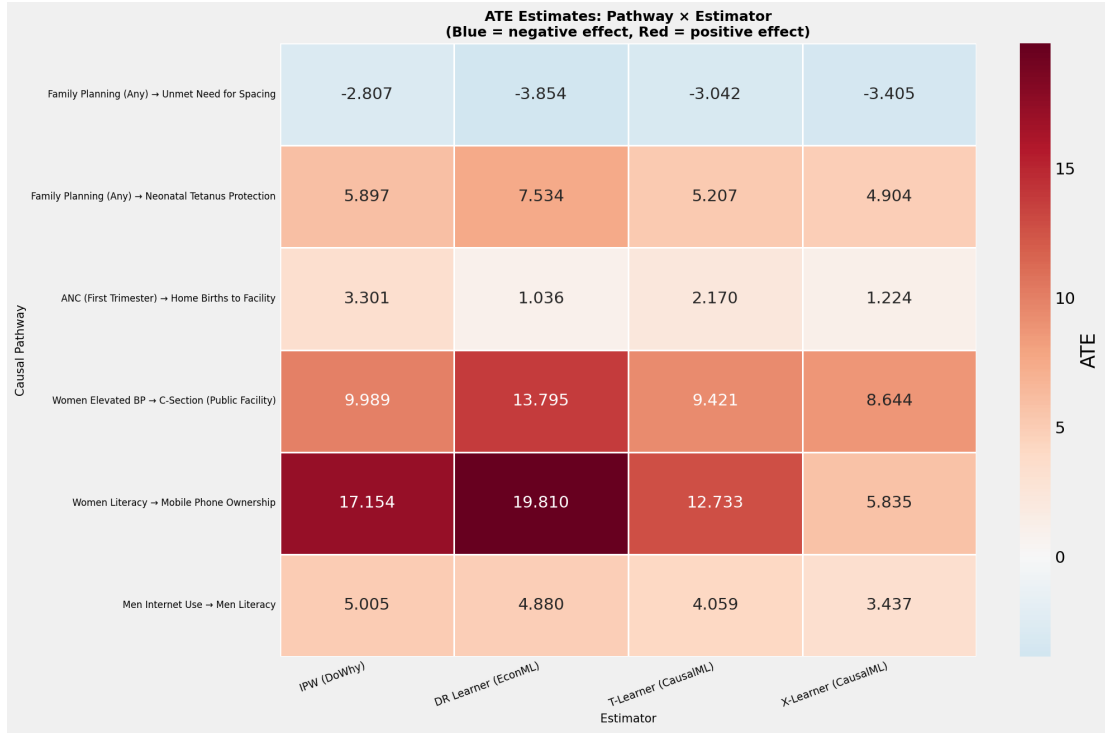


Figure I.1: ATE heatmap (pathway × estimator). Blue: negative; red: positive. Consistent direction across all four estimators visible for all six pathways.

Table I.3: Area-stratified ATE estimates (percentage points). *Consistent*: $\geq 3/4$ estimators agree on sign.

Pathway	Area	IPW	DR	T-Lrn	X-Lrn	Median	Cons.
FP → Spacing	Rural	−2.90	−3.50	−3.13	−3.23	−3.18	✓
	Urban	−3.06	−3.62	−2.76	−3.06	−3.06	✓
FP → Neonatal Tetanus	Rural	+4.84	+8.34	+6.27	+6.53	+6.40	✓
	Urban	+3.03	+3.40	+4.66	+6.16	+4.03	✓
ANC → Home Births	Rural	+2.67	+2.96	+1.52	+3.37	+2.81	✓
	Urban	−1.77	−2.14	−1.53	+0.43	−1.65	Partial
Elevated BP → C-Section	Rural	+9.49	+10.36	+6.07	+2.92	+7.78	✓
	Urban	+11.97	+17.34	+12.27	+11.49	+12.12	✓
Literacy → Mobile Phone	Rural	+18.74	+21.18	+18.85	+13.84	+18.80	✓
	Urban	+10.68	+10.62	+10.07	+8.58	+10.34	✓
Men Internet → Literacy	Rural	+6.40	+6.73	+5.09	+5.01	+5.74	✓
	Urban	+2.64	+3.97	+2.18	+2.10	+2.41	✓

Table I.4: Income-stratified ATE estimates (percentage points). GSDP: Low ($\lesssim$ Rs. 90k), Moderate, High ($\gtrsim$ Rs. 180k).

Pathway	GSDP	IPW	DR	T-Lrn	X-Lrn	Median	Cons.
FP $\rightarrow$ Spacing	Low	-2.85	-0.50	-5.55	-5.39	-4.12	✓
	Moderate	-3.53	-3.37	-3.29	-2.59	-3.33	✓
	High	-0.88	-1.05	-0.70	-0.46	-0.79	✓
FP $\rightarrow$ Neonatal TT	Low	+3.20	+1.20	+4.53	+4.45	+3.82	✓
	Moderate	+12.26	+14.08	+11.83	+12.67	+12.46	✓
	High	-0.93	+0.30	-0.65	-0.84	-0.74	Partial
ANC $\rightarrow$ Home Births	Low	+0.42	+2.60	+0.72	+0.74	+0.73	✓
	Moderate	+2.74	+9.37	+4.88	+5.34	+5.11	✓
	High	-2.51	-11.88	-3.14	—	-3.14	✓
BP $\rightarrow$ C-Section	Low	+9.89	+22.92	+5.88	+7.25	+8.57	✓
	Moderate	+10.94	+13.55	+11.01	+10.98	+10.99	✓
	High	+15.77	+14.37	+10.47	+8.85	+12.42	✓
Literacy $\rightarrow$ Mobile	Low	+13.71	+28.52	+12.80	+13.90	+13.81	✓
	Moderate	+20.18	+13.08	+16.54	+15.97	+16.26	✓
	High	+27.15	+27.92	+26.41	+23.04	+26.78	✓
Internet $\rightarrow$ Literacy	Low	+4.65	+15.98	+4.54	+4.84	+4.75	✓
	Moderate	+3.76	+8.12	+7.13	+8.87	+7.63	✓
	High	+4.46	+2.01	+4.87	+4.51	+4.49	✓

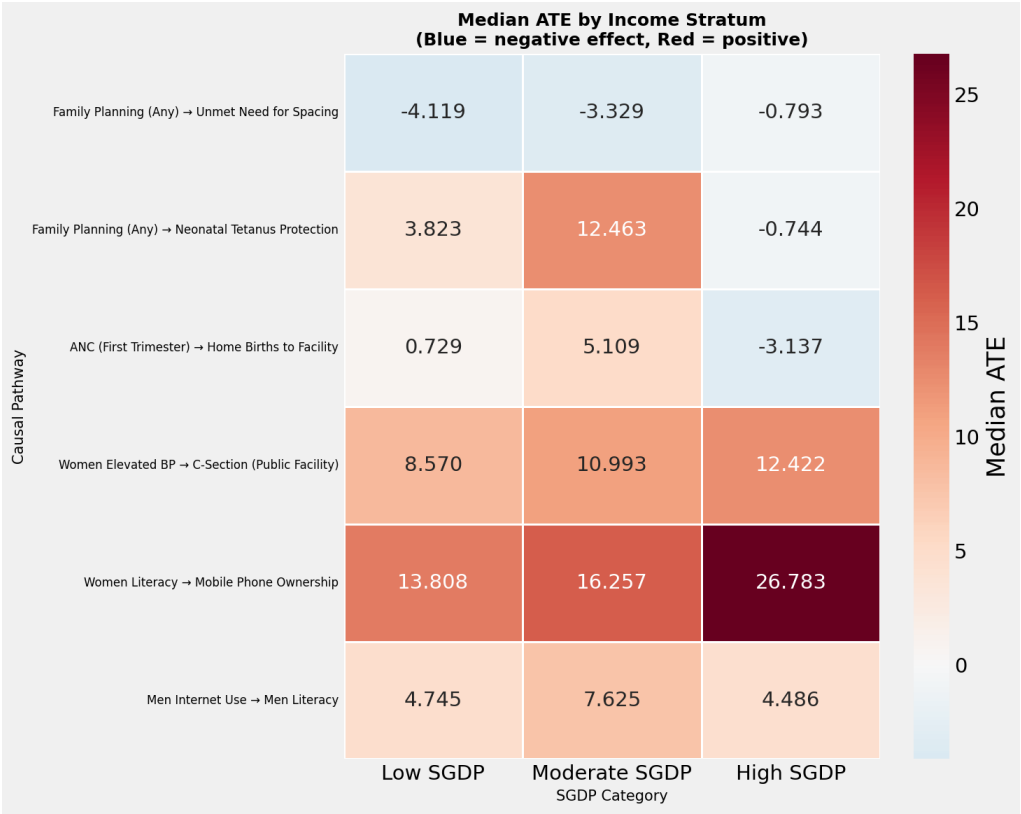


Figure I.2: Median ATE heatmap by pathway and income stratum. Blue: negative; red: positive.